

Computer Programming

Lecture 2 (Part I)

Outlines

- Problems
- Common Steps in Problem Solving
- Solution Representations
- Pseudocode
- Flowchart
- Solving sequential computational problems
- Selection

Problems

- Problems are something the result are not readily available.



Common Steps in problem solving

- Understanding the problem at hand precisely.
- Looking for possible similar or related solutions at hand. (Gathering information about the problem)
- Specifying possible of set of solutions
- Selecting the best possible solution
- Implementing and evaluating the solution
- Monitoring the solution to ensure it is effective

Representing a solution

- A solution to a problem needs to be carefully represented by outlining each individual steps of a solution precisely so that it can be understood, communicated, and reproduced by other interested parties.
- A computational solution can be represented:
 - Pseudocode: is a text-based detail design tool that allows the programmer to represent the implementation of an algorithm in a way that can be understood by anyone with basic programming knowledge.
 - Flowchart: is a diagrammatic representation of the steps of an algorithm.
 - Source code: a source code can be used to represent a computational solution using programming language.

Pseudocode

- Pseudocode is an informal language that helps programmers develop algorithms.
- Common instructions in a program:
 - Input: get data from keyboard, file, network and so on.
 - Output: displaying the output on screen, save it in a file, send it over the network, and so on.
 - Math: perform mathematical operations
 - Sequential execution: execute instructions in orderly manner
 - Conditional execution: check for certain condition and run the appropriate code.
 - Repetition: perform an action repeatedly

Exercise

Write a step by step instruction for cooking pasta with tomato sauce

Pseudocode

Goal: Computing hourly wage of an employee

Input hourly_rate, total_hours

Compute wage <- hourly_rate * total_hours

Output wage

Stop

Goal: Compute the area of a circle

Input radius

Compute area <- PI * radius * radius

Output area

Stop

Pseudocode Exercise

Write a pseudocode for the following computational problems:

- Compute circumference of a circle ($2 * \text{PI} * \text{radius}$)
- Compute the area of a rectangle
- Compute the average of four numbers

Flowchart

- Flowchart is a diagrammatic representation of the steps of an algorithm.
- Flowchart types:
 - Program flowchart: They contain the steps of solving a problem unit for a specific result.
 - System flowchart: contain the solutions of many problem units together that are closely related to each other and interact with each other to achieve a goal.

Program flowchart

- A program flowchart is an extremely useful tool in program development.
 - Any error or omission can be more easily detected from a program flowchart than it can be from a program because a program flowchart is a pictorial representation of the logic of a program.
 - Program flowchart can be followed easily and quickly.
 - It serves as a type of documentation, which may be of great help if the need for program modification arises in future.

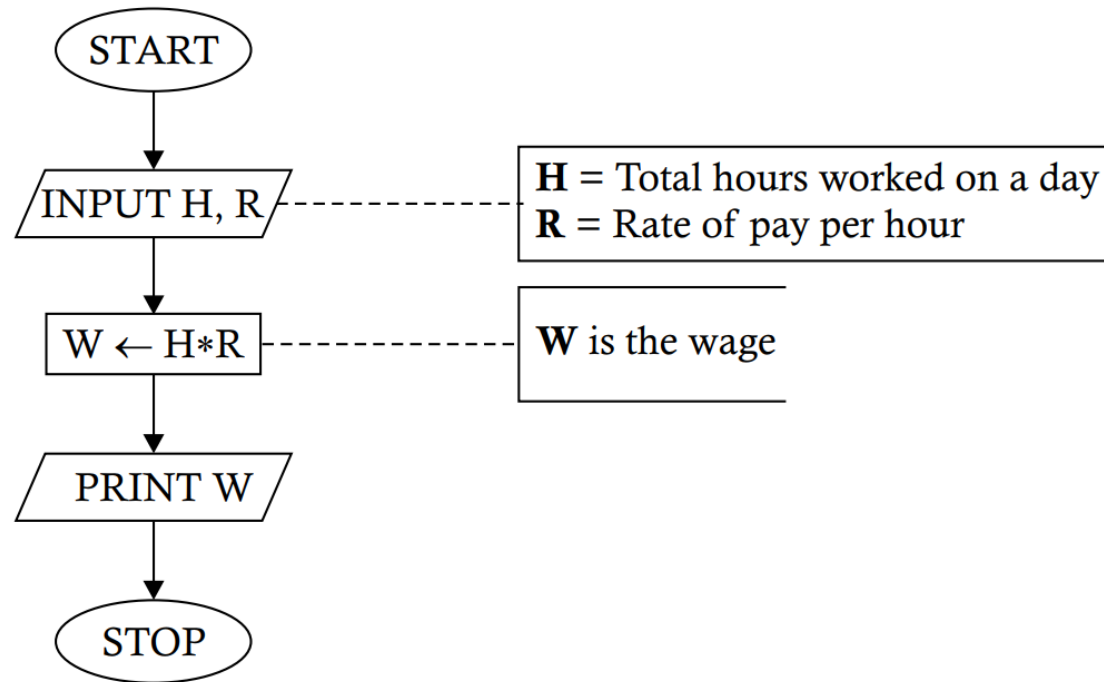
Flowchart Symbols

Symbol	Name and descriptions
	Terminal: used to show the beginning and end of a set of computer-related processes
	Input/Output: used to show any input/output operation
	Computer processing: used to show any processing performed by a computer system

Flowchart Symbols

Symbols	Name and descriptions
	Flow line: used to connect the symbols
	Comment: used to write any explanatory statement required to clarify something

Flowchart example



Goal: Computing hourly wage of an employe

Input hourly_rate, total_hours

Compute wage <- hourly_rate *
total_hours

Output wage

Stop

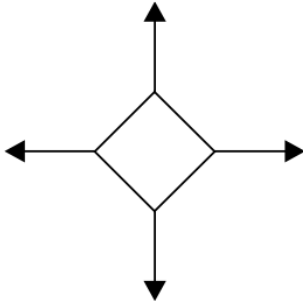
Flowchart Exercise

Draw a flowchart for the following computations:

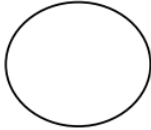
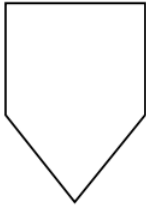
- Compute circumference of a circle ($2 * \text{PI} * \text{radius}$)
- Compute the area of a rectangle
- Compute the average of four numbers

The first five students to complete both exercises correctly get 1 bonus mark

More flowchart symbols

Symbols	Name and description
	Predefined processing: used to indicate any process not specially defined in the flowchart
	Document Input/Output: used when input comes from a document and output goes to a document.
	Decision: used to show any point in the process where a decision must be made to determine further action

More flowchart symbols

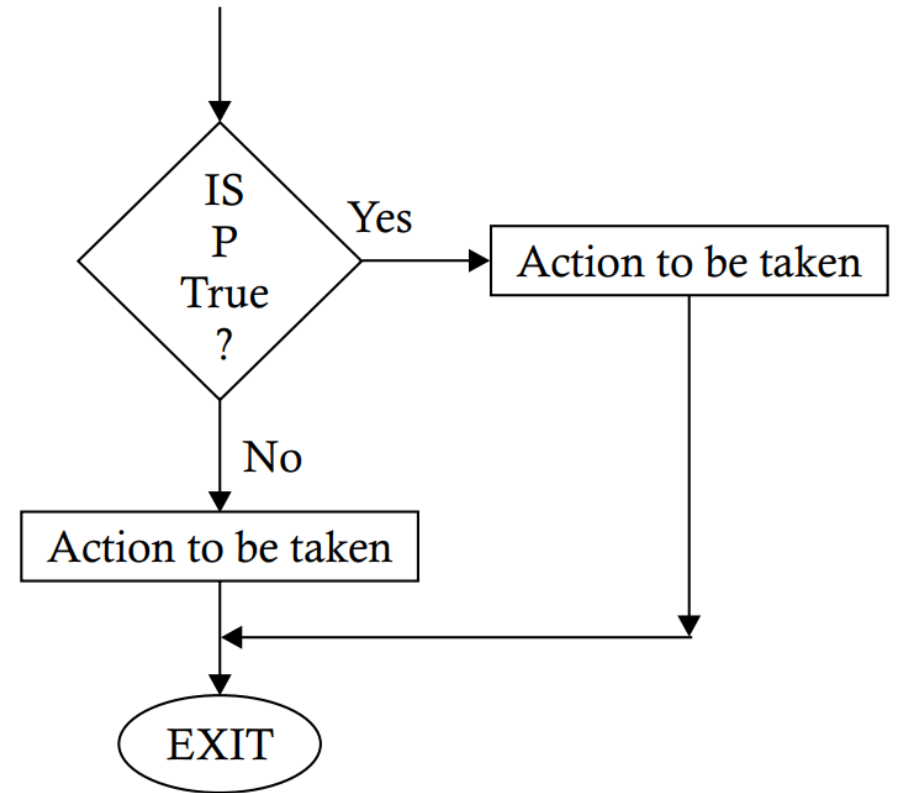
Symbols	Name and description
	On-page connector: used to connect parts of a flowchart continued on the same page
	Off-page connector: used to connect parts of a flowchart continued to separate pages

Flowchart rules

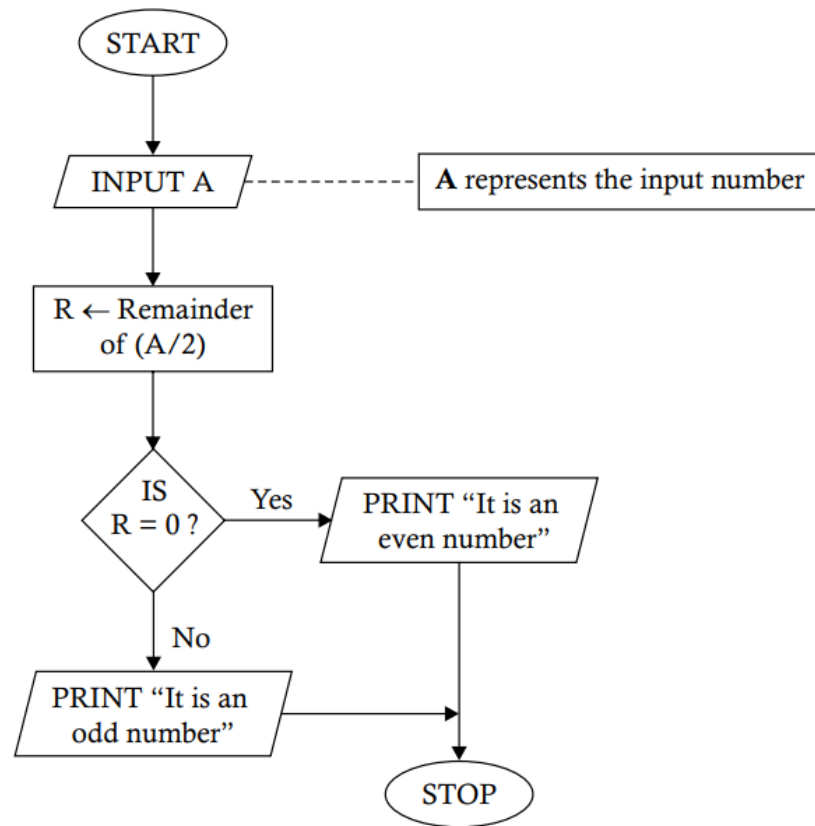
- Only the standard symbols should be used in program flowcharts.
- The program logic should depict the flow from top to bottom and from left to right.
- Each symbol used in a program flowchart should contain only one entry point and one exit point, with the exception of the decision symbol. This is known as the **single rule**.
- The operations shown within a symbol of a program flowchart should be expressed independently of any particular programming language.
- All decision branches should be well-labeled.

Selection

- This process of decision-making is implemented through a logic structure called selection.
- A predicate (condition) is tested, if it is true or false.
 - If true, a course of action will be specified for it.
 - If false, an alternative course of action will be specified for it



Selection



Goal: Determine if number is odd or even

Input A

Compute $R \leftarrow \text{Remainder of } (A/2)$

If $R = 0$

Then Print "It is an even number"

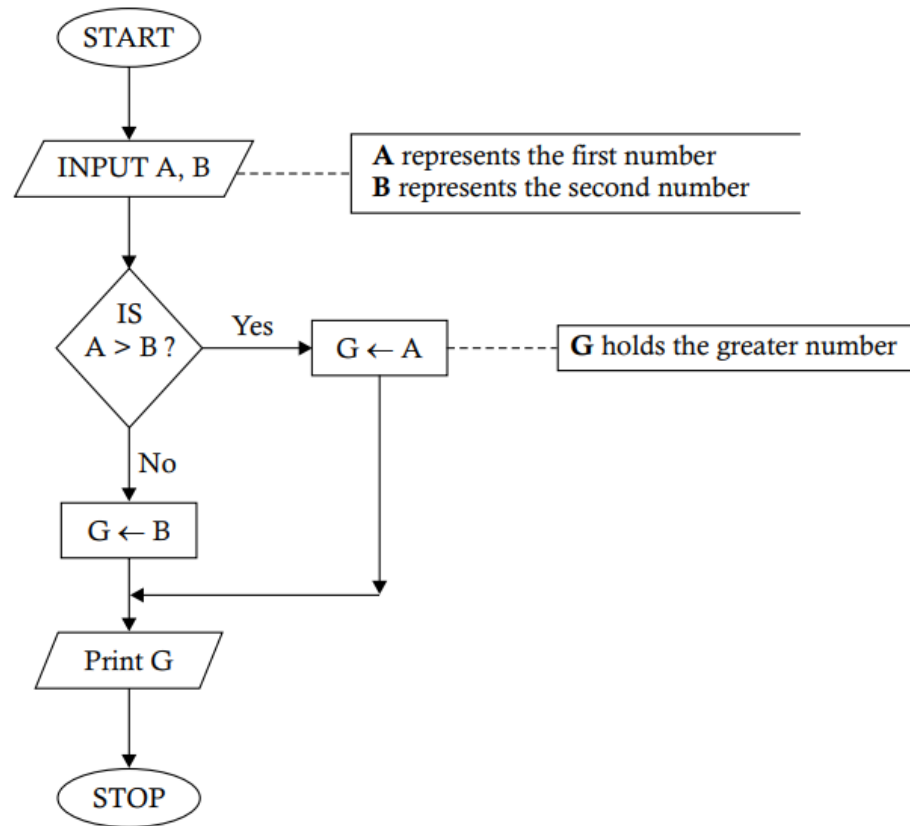
Else

Print "It is an odd number"

End-If

Stop

Selection



Goal: Output the maximum of two numbers

Input A,B

If $A > B$

Then $G \leftarrow A$

Else

$G \leftarrow B$

End-If

Print G

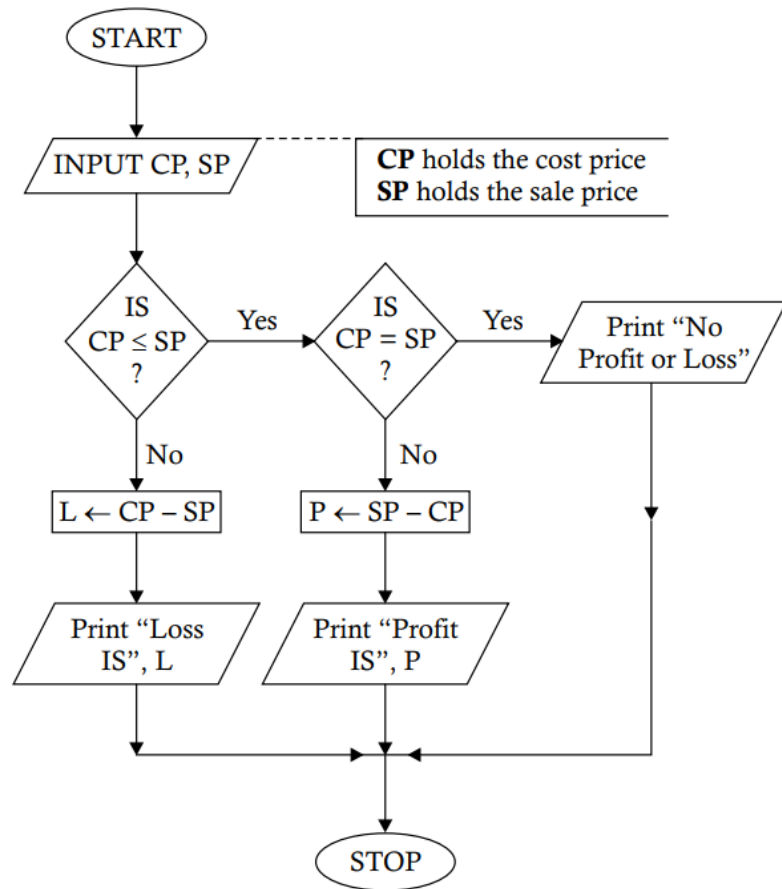
Stop

Selection Exercise

- Write a pseudocode for a computation that accepts two numbers and tells if the first number is a multiple of the second number
- Draw a flowchart for the pseudocode

The first 5 students to return correct answer for the questions get 1 bonus mark

Selection



Goal: Determine loss or profit

Input CP,SP

If $CP \leq SP$

Then

If $CP = SP$

Then print "No profit or loss"

Else

Compute $P \leftarrow SP - CP$

print "The profit is ", p

End-if

Else

Compute $L \leftarrow CP - SP$

print "The loss is ", L

End-if

Stop

More Selection Exercise

- Write a pseudocode to determine the maximum of 3 numbers
- Draw a flowchart for the pseudocode

The first 5 students to complete the exercise with a correct answer get bonus 1 mark